EXPERIMENT 3

DECAY OF CURRENT IN A CAPACITIVE CIRCUIT

AIM: To study decay of current in a capacitive circuit and determine its time constant.

APPARATUS: Resistor, Capacitor, Constant voltage source (battery/battery eliminator), Multi-meter, Keys, Stopwatch, Connecting wires, Bread board.

THEORY: A capacitance can be used to store energy by accumulation of charges and a charged capacitor can be used to deliver energy by discharging it through a load. Both charging and discharging are not instantaneous. It can be shown that the charging and discharging times depend on the capacitance and total resistance of the circuit.

(a) Charging:

For the circuit in Fig. 1, having capacitance C and resistance R, the equation for potential with key K_1 closed and K_2 open, is

$$V_0 = iR + \frac{q}{C} \quad \text{.....(1)}$$

Where, i is the current in the circuit and q is the charge accumulated in the capacitor. Since, $i = dq/dt$, we get the equation for q as

$$V_0 = R\frac{dq}{dt} + \frac{q}{C} \quad \text{.....(2)}$$

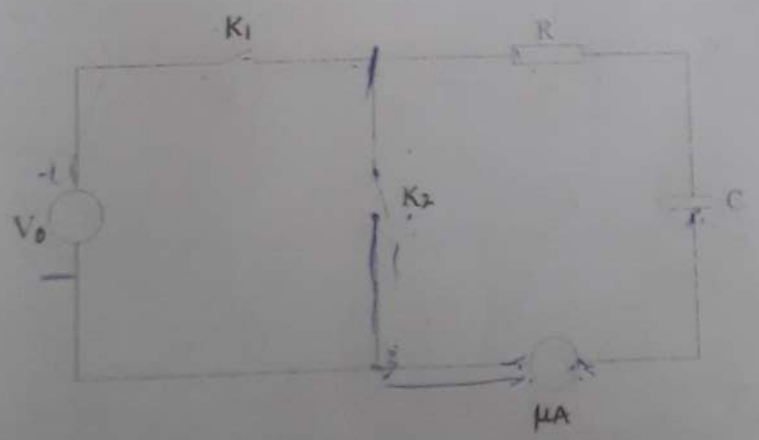

Figure 1: Circuit Diagram

The solution of this equation is

$$q = CV_0(1 - e^{-t/\tau}) \quad \text{.....} (3)$$

Where, $\tau = RC$ is the time constant of the circuit. The current i is given by

$$\boxed{i = \frac{dq}{dt} = \frac{V_0}{R} e^{-t/\tau}} \quad \text{.....} (4)$$

The voltage across C will be

$$V = V_0 (1 - e^{-t/\tau}) \quad \text{.....} (5)$$

Equations **(4)** and **(5)** describe the decay of current in the circuit and growth of voltage in the capacitor respectively, during charging of the capacitor respectively.

(b) Discharging:

When key K_1 is open and K_2 is closed, with the capacitor charged to a voltage V_0, discharging of the capacitor through the resistance R is described by the following equations

$$q = CV_0 e^{-t/\tau} \quad \text{.....} (6)$$

$$i = \frac{dq}{dt} = -\frac{V_0}{R} e^{-t/\tau} \quad \text{.....} (7)$$

$$V = V_0 e^{-t/\tau} \quad \text{.....} (8)$$

Equations **(7)** and **(8)** describe the decay of current in the circuit and decay of voltage across C respectively, during the discharging process.

Determination of :

From equation (4) which gives current in the circuit during charging, we get

$$\ln i = -\frac{1}{\tau} t + \ln\left(\frac{V_0}{R}\right) \qquad (9)$$

Thus the relation between **ln*i*** and **t** is a straight line with slope equal to **-1/ τ**. Therefore time constant, Can be determined by calculating the slope of the **ln*i*** versus **t** plot. Similarly, **τ** can also be determined using equation **(7)** which gives current in the circuit during discharging

PROCEDURE:

(a) Charging:

1. Assembly the circuit as shown in figure 1.
2. Set the source voltage V_0 at a value between **2** and **4** Volts. This voltage should be kept constant through the experiment.
3. Set the ammeter (if adjustable) in the appropriate current range. Magnitude of current can be estimated from the value of V_0 and R.
4. Discharge the capacitor by short circuiting its two leads.
5. Complete the charging circuit by closing the key $\mathbf{K_1}$ (Keeping $\mathbf{K_2}$ open) and simultaneously start a stop watch.
6. Measure current i as a function of time t (Table) at convenient interval (say **10** seconds). Continued to take readings until the current appears to become constant with time.

(b) Discharging:

(1). After charging C to full voltage, discharge it through R by opening the key $\mathbf{K_1}$ and then closing key $\mathbf{K_2}$. Simultaneously start a stopwatch.

(2). Repeat step (a) – 6.

(c) Determine of τ:

1. Plot the **ln*i*** vs **t** graphs for both the discharging data.

2. Calculate the slopes and from it calculate the value of τ for both the two cases. Take average between the two values of τ.

3. Calculate the theoretical value of τ (= RC) and compare with the experimental obtained value.

OBSERVATIONS:

$V_0 =$

$R =$

$C =$

(a) **Charging:**

Sl. No.	Time(t) Sec	Current (i) amp	ln*i*	Slope = $1/\tau_1$	Time constant τ_1

(b) Discharging:

Sl. No.	Time(t) Sec	Current *(i)* amp	ln*i*	Slope = $1/\tau_2$	Time constant τ_2

CALCULATIONS:

Theoretical value of $\tau = RC =$.....

Practical value $= \frac{\tau_1 + \tau_2}{2} =$ .....

RESULT:

Theoretically the value of τ, time constant in a capacitive circuit is found to be .....,

and practically the value of τ, time constant in a capacitive circuit is found to be .....

PRECAUTIONS:

1. It should be ensured that the capacitor is fully discharged before starting charging and that it is fully charged before discharging.
2. Keys K_1 and K_2, should not be closed simultaneously.